

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	21.0000	44.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.0000	-0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.4640	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	55.8690	117.0600
Tip diameter (mm)	d_a	60.869 (60.869 / 60.769)	122.060 (122.060 / 121.960)
Root diameter (mm)	d_f	49.619 (49.427 / 49.317)	110.810 (110.617 / 110.507)
Number of teeth spanned	k	3.0000	6.0000
Base tangent length (no backlash) (mm)	W_k	19.329 (19.263 / 19.225)	42.431 (42.365 / 42.328)
Effective Diameter of ball/pin (mm)	D_M	4.5000	4.2500
Diametral measurement over two balls (mm)	M_{dk}	62.333 (62.175 / 62.083)	122.879 (122.701 / 122.598)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.2040		1.2316		1.0000
Bending S.F	3.0900		3.2725		1.4000
Scuffing S.F (Flash Temp.)	2.3658				2.0000
Micropitting S.F	0.7782				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.2000	0.0000	0.1000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4628
Overlap Ratio	1.9596
Total Contact Ratio	3.4224
Worst Pitch Line Velocity [m/s]	29.2531
Total Efficiency [%]	98.6671
Worst Mesh Loss [W]	784.5186
Worst Churning Loss [W]	1214.8574
Worst Peak-to-Peak T.E. [μm]	0.2009
Gear Total Mass [kg]	1.7826
Hunting tooth	Complete

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	20.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	21.0000	44.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.0000	-0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	86.4644	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9600	0.9600
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.0952	
Transverse module (mm)	m_t	2.6604	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.1728	
Working transverse pressure angle (deg)	α_{wt}	21.1728	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.1916 / 21.1540	
Base helix angle (deg)	β_b	18.7472	
Helix angle at operating pitch circle (deg)	β_w	20.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Circumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0745 / 0.1171	0.0745 / 0.1171
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1405 / 0.2426	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1240 / 0.2143	
Angular backlash, min (deg)	j_θ, min	0.2881	0.1375
Angular backlash, max (deg)	j_θ, max	0.4977	0.2375
Effective facewidth (mm)	b_{eff}	45.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0385 / -0.0604	-0.0385 / -0.0604
Prevent undercut profile shift coefficient	x_{Emin}	-0.4577	-2.0542
Max. profile shift coeff. (min top land)	x_{Emax}	1.1580	1.9311
Reference center distance, $x_1=x_2=0$ (mm)	a_d	86.4644	
Working center distance (mm)	a_w	86.4644	
Reference diameter (mm)	d	55.8693	117.0596
Operating pitch diameter (mm)	d_w	55.8693	117.0596
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	55.8764 / 55.8622	117.0744 / 117.0447
27		60.8693	122.0596
Tip diameter with tolerance (mm)	$d_a, e/i$	60.8693 / 60.7693	122.0596 / 121.9596

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	49.6193	110.8096
Root diameter with tolerance (mm)	d_f, e/i	49.4270 / 49.3171	110.6172 / 110.5073
Tip form diameter (mm)	d_Fa	60.6693	121.8596
Tip form diameter with tolerance (mm)	d_Fa, e/i	60.4518 / 60.3279	121.6536 / 121.5362
Root form diamter (mm)	d_Ff	52.4818	112.8008
Root form diameter with tolerance (mm)	d_Ff, e/i	52.4201 / 52.3873	112.6677 / 112.5927
Active tip diameter (EAP) (mm)	d_Na	60.6693	121.8596
Active tip diameter with tolerance (mm)	d_Na, e/i	60.4518 / 60.3279	121.6536 / 121.5362
Active root diameter (SAP) (mm)	d_Nf	52.7531	113.5754
Active root diameter with tolerance (mm)	d_Nf, e/i	52.8834 / 52.8181	113.7799 / 113.6769
Mean diameter for mass calculation (mm)	d_m	55.0957	116.2859
Gear mass (kg)	m	0.5042	1.2783
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.6250	0.6250
Effective tip clearance, e/i (mm)	c, e/i	0.8371 / 0.7102	0.8371 / 0.7102
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.1990	0.5046

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.7761 / 5.6712
Working depth (mm)	h_w	5.0000	
Addendum (mm)	h_a	2.5000	2.5000
Addendum, e/i (mm)	h_a, e/i	2.5000 / 2.4500	2.5000 / 2.4500
Dedendum (mm)	h_f	3.1250	3.1250
Dedendum, e/i (mm)	h_f, e/i	3.2212 / 3.2761	3.2212 / 3.2761
Number of virtual gear teeth	z_n	24.9220	52.2176
Tooth thickness coeff.	s_nc	1.5708	1.5708
Transverse tooth thickness (mm)	s_t	4.1790	4.1790
Normal tooth thickness (mm)	s_n	3.9270	3.9270
Normal tooth thickness at d_a (mm)	s_an	1.8045	1.9478
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.7826 / 1.6859	1.9205 / 1.8336
Normal tooth thickness at d_Fa (mm)	s_Fan	1.9109	2.0381
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	2.0137 / 1.9068	2.1099 / 2.0164
Normal space width at d_f (mm)	e_fn	0.0000	2.0587
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.0912 / 2.1107

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.8540	
Transverse pitch (mm)	p_t	8.3580	
Normal base pitch (mm)	p_bn	7.3803	
Transverse base pitch (mm)	p_bt	7.7938	
Axial pitch (mm)	p_x	22.9635	
Lead (mm)	p_z	482.2336	1010.3943

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	11.4005	
Length of path of contact, e/i (mm)	g_α, e/i	10.9852 / 10.6686	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.9451	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.7333 / 5.6921	
Length of recess POC of pinion (C to E) (mm)	g_a1	5.4554	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	5.2519 / 5.2323	
Length T1 - A (mm)	T1A	4.1444	
T1A tolerance (mm)	e/m/i	4.5406 / 4.4435 / 4.3464	
Length T1 - B (mm)	T1B	7.7511	
T1B tolerance (mm)	e/m/i	7.5378 / 7.4765 / 7.4153	
Length T1 - C (mm)	T1C	10.0895	
T1C tolerance (mm)	e/m/i	10.0993 / 10.0895 / 10.0797	
Length T1 - D (mm)	T1D	11.9382	
T1D tolerance (mm)	e/m/i	12.3344 / 12.2373 / 12.1402	
Length T1 - E (mm)	T1E	15.5449	
T1E tolerance (mm)	e/m/i	15.3316 / 15.2704 / 15.2091	
Length T2 - A (mm)	T2A	27.0850	
T2A tolerance (mm)	e/m/i	26.8526 / 26.7860 / 26.7193	
Length T2 - B (mm)	T2B	23.4783	
T2B tolerance (mm)	e/m/i	23.8446 / 23.7529 / 23.6612	
Length T2 - C (mm)	T2C	21.1399	
T2C tolerance (mm)	e/m/i	21.1605 / 21.1399 / 21.1193	
Length T2 - D (mm)	T2D	19.2912	
T2D tolerance (mm)	e/m/i	19.0588 / 18.9921 / 18.9255	
Length T2 - E (mm)	T2E	15.6845	
T2E tolerance (mm)	e/m/i	16.0507 / 15.9591 / 15.8674	
Length T1 - T2 (mm)	T1T2	31.2294	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	31.2599 / 31.2294 / 31.1990	
Diameter of single contact point A (mm)	d_A	52.7531	121.8596
Diameter of single contact point B (mm)	d_B	54.3554	118.8288
Diameter of single contact point C (mm)	d_C	55.8693	117.0596
Diameter of single contact point D (mm)	d_D	57.3086	115.7755
Diameter of single contact point E (mm)	d_E	60.6693	113.5754
Transverse contact ratio	ϵ_α	1.4628	
Overlap ratio	ϵ_β	1.9596	
Total contact ratio	ϵ_γ	3.4224	
Total contact ratio with tolerance	e/m/i	3.369 / 3.349 / 3.328	
Specific sliding at Tip	ζ_a	0.5184	0.6794
Specific sliding at Root	ζ_f	-2.1191	-1.0766

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	19.0000
Profile form deviation (μm)	f_fa	9.0000	9.0000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	28.0000
Runout (μm)	F_r	22.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	33.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	3.0000	6.0000
Base tangent length (mm)	W_k	19.3286	42.4310
Base tangent length with tolerance (mm)	W_k, e/i	19.2629 / 19.2253	42.3653 / 42.3277
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	55.1930	116.2899
Theoretical diameter of ball/pin (mm)	D_M.theory	4.2792	4.2249
Effective Diameter of ball/pin (mm)	D_M.eff	4.5000	4.2500
Radial single ball measurement (mm)	M_rk	31.2476	61.4396
Radial single ball measurement with tolerance (mm)	M_rk, e/i	31.1681 / 31.1223	61.3505 / 61.2992
Diametral two ball measurement (mm)	M_dk	62.3331	122.8791
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	62.1745 / 62.0831	122.7010 / 122.5985
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1586 / -0.2500	-0.1782 / -0.2806

3. Detail rating

3.1. General

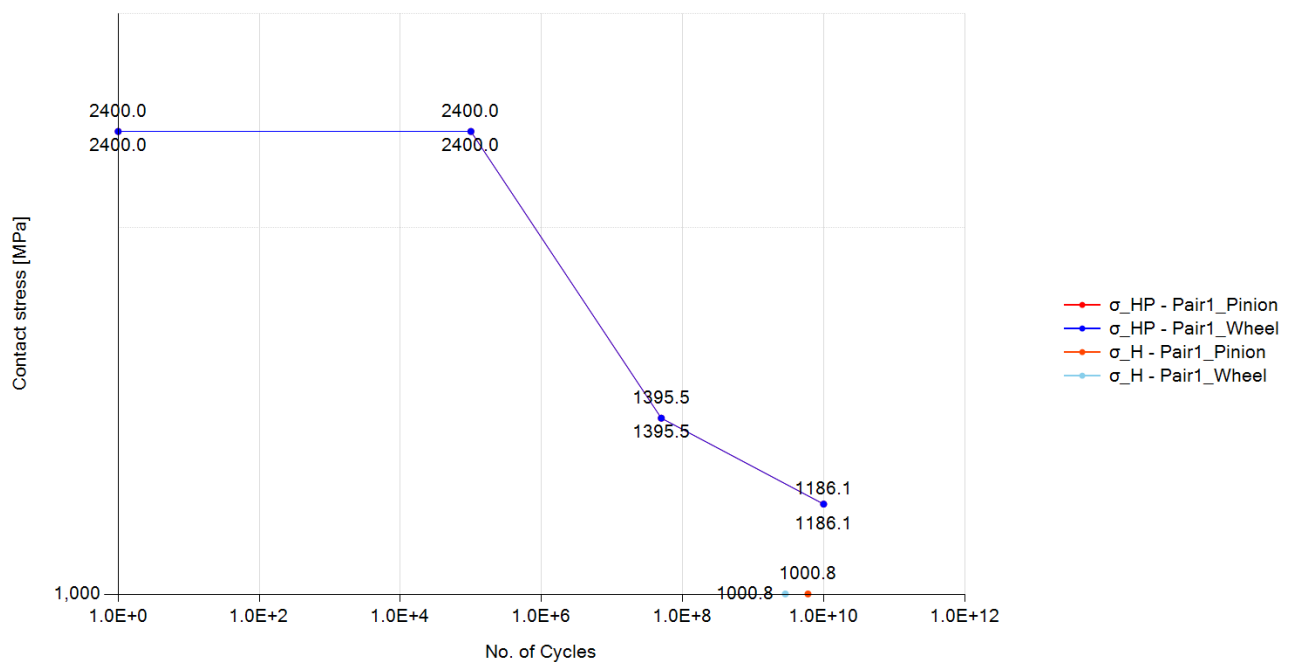
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4772.7270
Nominal tangential load at d (N)	F_t	5127.6590	
Nominal axial load at d (N)	F_a	1866.3150	
Nominal radial load at d (N)	F_r	1986.0910	
Nominal normal load at d (N)	F_n	5806.9420	
Nominal tangential load per unit facewidth (N/mm)	w	113.9480	
Circumferential speed at d (m/s)	v	29.2530	
Sliding velocity at Tip dia. (m/s)	v_g	8.4400	9.1970
Nominal tangential load at d_w (N)	F_{tw}	5127.6590	
Nominal axial load at d_w (N)	F_{aw}	1866.3150	
Nominal radial load at d_w (N)	F_{rw}	1986.0910	
Circumferential speed at d_w (m/s)	v_w	29.2530	
Running-in value (μm)	y_p	0.5930	
Running-in value (μm)	y_f	0.6750	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	17.1210	
Single tooth stiffness (N/mm/ μm)	c'	11.2690	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	15.1800	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	12.9030	
Mean diameter for K_v (mm)	$d_m(K_v)$	55.2440	116.4350
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J(K_v)$	309.4920	4482.2260
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	6.8780	99.6050
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0101	0.0334
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0078	
Resonance running speed (rpm)	n_{E1}	20088.6890	
Resonance ratio	N	0.4980	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_v	1.1780	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2580	
Transverse load factor - flank	$K_{H\alpha}$	1.1790	
Transverse load factor - root	$K_{F\alpha}$	1.1790	
Number of load cycles	N_L	6.0000×10^9	2.8640×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



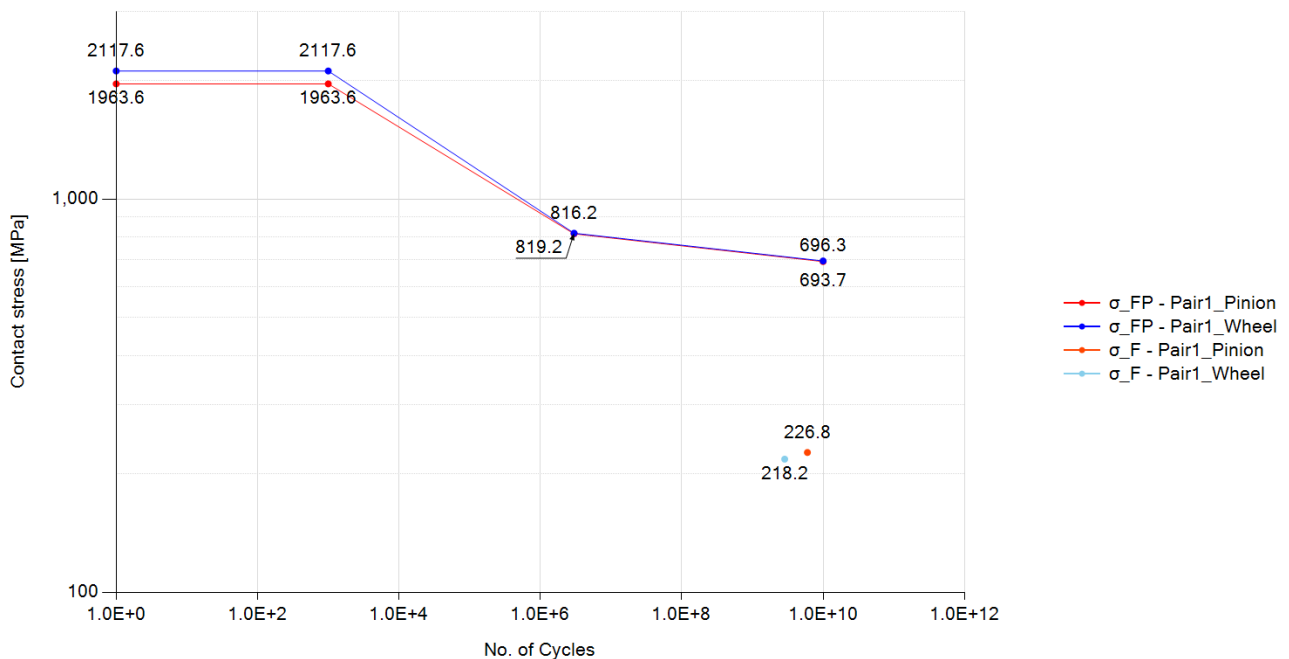
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.8270	
Zone factor	Z_H	2.3710	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0320	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	666.3940	
Contact stress (N/mm ²)	σ_H	1000.7650	1000.7650
Life factor	Z_{NT}	0.8630	0.8830

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0320	1.0320
Roughness factor	Z_R	0.9530	0.9530
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1395.4570	1395.4570
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1186.1390	1186.1390
Permissible contact stress (N/mm ²)	σ_{HP}	1204.8710	1232.5200
Safety factor for Contact	S_H	1.2040	1.2320
Contact damage (%)	-	0.2360	0.1120
Cycle to failure	-	2.5470×10^{12}	2.5470×10^{12}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.2590	1.1110
Stress correction factor	Y_S	1.8030	1.9660

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.6610	19.8520
Bending moment arm (mm)	h_{Fe}	2.6180	2.7580
Tooth thickness at root (mm)	s_{Fn}	4.9610	5.4020
Tooth root radius (mm)	ρ_F	1.4540	1.3170
Helix angle factor	Y_β	1.0040	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	103.9360	99.9720
Tooth root stress (N/mm ²)	σ_F	226.8200	218.1700
Life factor	Y_{NT}	0.8590	0.8720
Relative notch sensitive factor	$Y_{\delta relT}$	0.9920	0.9960
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	1963.5900	2117.5710
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	816.1590	819.1620
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	693.7350	696.2870
Permissible tooth root stress (N/mm ²)	σ_{FP}	700.8710	713.9520
Safety factor for Bending	S_F	3.0900	3.2720
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	8.6990×10^{26}	7.2770×10^{27}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	38.2500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	256.9860	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	9.3830	
Angle factor	X_αβ	0.9690	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0460	
Maximum flash temperature (deg)	θ_fl_max	101.7020	
Mean flash temperature (deg)	θ_fl_mean	33.1470	
Total mass temperature (deg)	θ_M	98.6950	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	200.3970	
Safety factor for flash temperature	S_B	2.3660	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0986
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.1643
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.7782
transverse tangential load at reference cylinder per mesh (N)	F_t	5127.6590
transverse load in plane of action (N)	F_bt	5498.8590
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1820
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.3000
load losses factor	H_v	0.1420
mean coefficient of friction	μ_m	0.0560
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8890
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	17.8770
sliding velocity at point A (m/s)	v_g,A	-9.1970
tangential velocity on pinion at point A (m/s)	v_r1,A	4.3400
tangential velocity on wheel at point A (m/s)	v_r2,A	13.5370
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	101.7560
flash temperature at point A (°C)	θ_fl,A	124.4430
contact temperature at point A (°C)	θ_B,A	226.1990
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3200×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	6.8870×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	1.5020
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0010
normal radius of relative curvature at point A (mm)	ρ_n,A	3.7960
material parameter	G_M	2988.1700
local velocity parameter at point A	U_A	7.2440×10 ⁻¹¹
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.0830
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	1015.2820
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1524.7120

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.0952
Tangential speed at working pitch dia. (m/s)	v_{wt}	29.2531
Tangential force at pitch dia. (N)	F_t	5127.6592
Tooth normal force, transverse section (N)	F_{bt}	5498.8587
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	21.1314
Equivalent radius of curvature (mm)	ρ	7.2125
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0368
Tooth loss factor	H_v	0.1421
Mesh power loss (W)	P_{VZP}	784.5186
Mesh efficiency (%)	η_z	99.4770

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4772.7273
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	27.9347	58.5298
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	88584.9784	88584.9784
Froude number	Fr	3122.7105	1490.3845
Centrifugal acceleration (m/s ²)	γ	16062.8489	4682.0237

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	60.8693	122.0596
Pitch diameter of gear (mm)	d_p	55.8693	117.0596
Facewidth (mm)	b	45.0000	45.0000
Dimensionless drag torque	C_m	0.0013	0.0029
Immersion area of the gear (m^2)	S_m	0.0124	0.0316
Total oil churning loss (W)	P	182.2878	1032.5696
Total oil churning torque (N.m)	T	0.1741	2.0660

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000